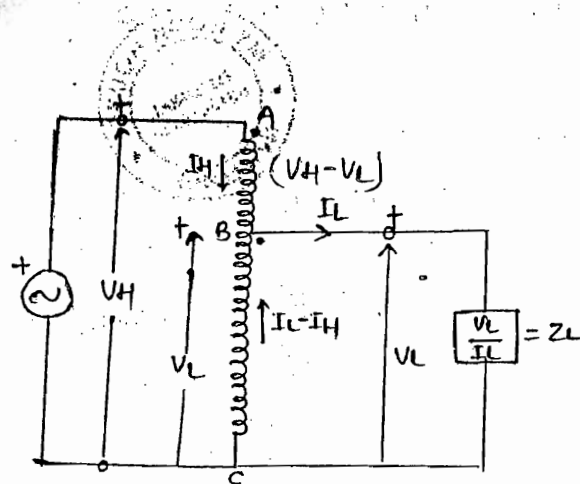


Auto Transformer



Since the flux is common;

$$\frac{V_H}{N_H} = \frac{V_L}{N_L}$$

$$\frac{V_H}{V_L} = \frac{N_H}{N_L} = a(\text{auto}) = \frac{I_L}{I_H}$$

provided $\rightarrow$

$$I_L - I_H < I_H$$

Total wdg (AC) : N_H

Common wdg (BC) : N_L

Series wdg (AB) : $N_H - N_L$

$$\therefore I_L - I_H < I_H$$

$$I_L < 2I_H$$

$$\frac{I_L}{I_H} < 2$$

$$a < 2$$

$$a(\text{auto}) < 2$$

mmf balance (neglecting I_0)

$$(N_H - N_L) I_H = N_L (I_L - I_H)$$

$$N_H I_H - N_L I_H = N_L I_L - N_L I_H$$

$$N_H I_H = N_L I_L$$

$$\boxed{\frac{N_H}{N_L} = \frac{I_L}{I_H}}$$

220V, 100W

110V; $P = ?$

$$P = \frac{V^2}{R}$$

$$P \propto V^2$$

$$\frac{1}{4} \times 100 = 25W$$

Copper Comparison $\rightarrow$

Copper Weight = Copper Volume $\times$ Copper density

$\propto$ Volume

$\propto$ Cond^rc/s $\times$ Cond^rlength

$\downarrow$

$\propto I$

$\downarrow$

$\propto N$

$\propto NI$

$\propto$ mmf

$$\begin{aligned}
 \therefore \frac{Cu(\text{auto})}{Cu(\text{2wdg})} &= \frac{(N_H - N_L) I_H + N_L (I_L - I_H)}{N_H I_H + N_L I_L} \\
 &= \frac{N_H I_H - N_L I_H + N_L I_L - N_L I_H}{2 N_H I_H} \\
 &= \frac{2 N_H I_H - 2 N_L I_H}{2 N_H I_H} \\
 &= \frac{N_H - N_L}{N_H} \\
 &= 1 - \frac{N_L}{N_H}
 \end{aligned}$$

$$\boxed{\frac{Cu(\text{auto})}{Cu(\text{2wdg})} = \left[1 - \frac{1}{a(\text{auto})} \right]}$$

$$\begin{aligned}
 \text{Cu saving} &= \frac{Cu(\text{2wdg}) - Cu(\text{auto})}{Cu(\text{2wdg})} \\
 &= 1 - \frac{Cu(\text{auto})}{Cu(\text{2wdg})} \\
 &= 1 - \left[1 - \frac{1}{a(\text{auto})} \right]
 \end{aligned}$$

$$\boxed{\% \text{Cu saving} = \frac{1}{a(\text{auto})} \times 100\%}$$

Components of power TF $\rightarrow$

$$\begin{aligned}
 S_L &= V_L I_L \\
 &= V_L [(I_L - I_H) + I_H] \\
 &= V_L (I_L - I_H) + V_L I_H \\
 &\quad \downarrow \quad \quad \quad \downarrow \\
 &\text{Inductive Xfer} \quad \text{Conductive Xfer}
 \end{aligned}$$

Inductive Xtransfer = A transfer due to transformer action.

$$\begin{aligned}
 S_{BC} &= V_L(I_L - I_H) \\
 &= V_L I_L - V_L I_H \\
 &= V_H I_H - V_L I_H \\
 &= (V_H - V_L) I_H
 \end{aligned}$$

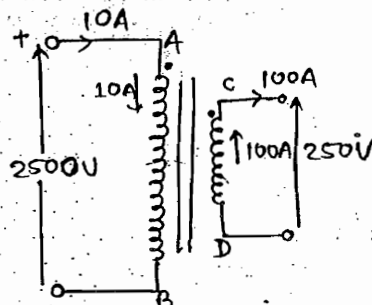
$$S_{BC} = S_{AB}$$

Ratio →

$$\begin{aligned}
 \frac{S_{cond}}{S_{total}} &= \frac{V_L I_H}{V_L I_L} \\
 &= \frac{I_H}{I_L} \\
 &= \frac{1}{q_{auto}} = cu(\text{saving})
 \end{aligned}$$

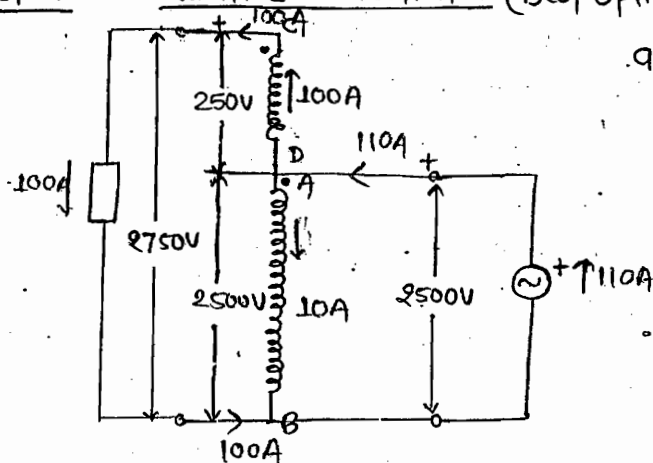
$$\frac{S_{und}}{S_{total}} = 1 - \frac{1}{q_{auto}}$$

Que. → A 25 kVA, 2500/250V 2wdg TF is to be reconnected as auto TF.



Determine the vol. Ratio & the kVA rating as auto TF for all possible connection.

Soln → additive connection → (Best option)

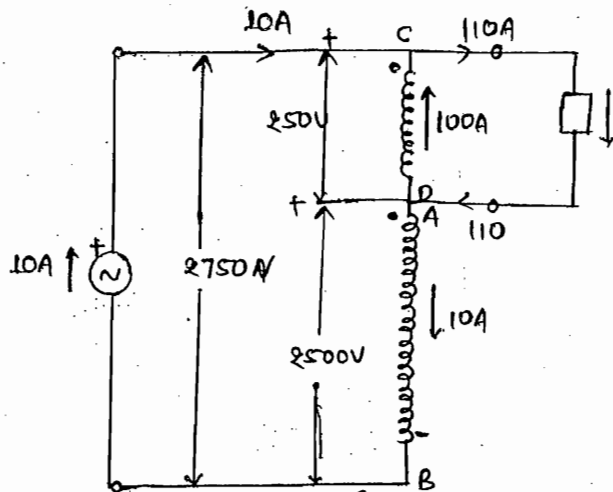


$$q_{auto} = \frac{2750}{2500} = 1.1$$

$$\begin{aligned}
 S_{auto} &= 2750 \times 100 \quad (\text{OR}) \quad 2500 \times 110 \\
 &= 275 \text{ kVA}
 \end{aligned}$$

Given → 25 kVA 250
 275 kVA
 ↓
 Inductive Capacitive
 tra Conductive

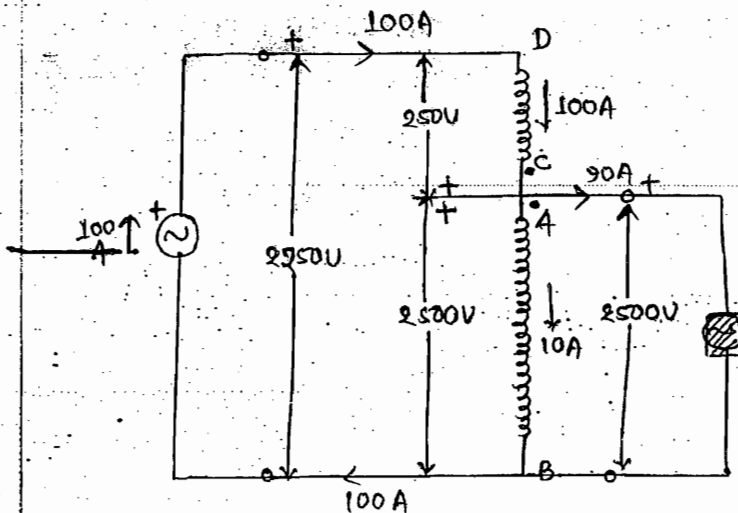
(Bad option)



$$a_{(auto)} = \frac{2750}{250} = 11$$

$$S_{auto} = 2750 \times 10 \text{ (OR)} 250 \times 110 \\ = 27.5 \text{ kVA}$$

Subtractive Connection → (Good option)



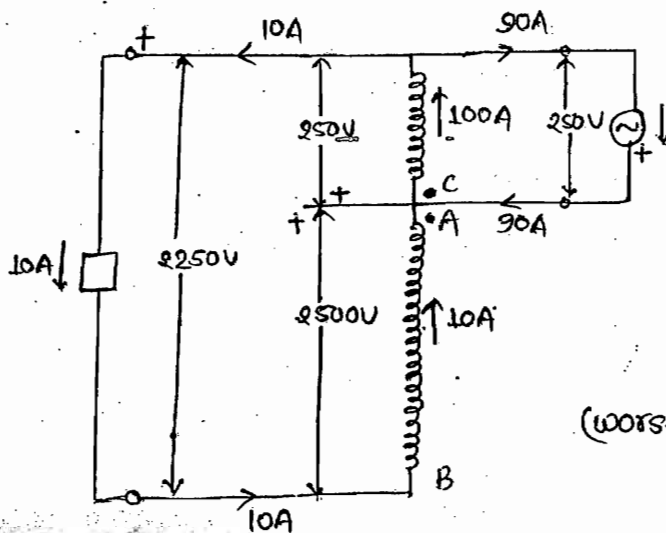
$$a_{(auto)} = \frac{2500}{250} = 1.111$$

$$S_{auto} = 2500 \times 90 \text{ OR } 250 \times 100 \\ = 22.5 \text{ kVA}$$

Conductive = 200 kVA

Inductive X_{fer} = Common X_{fer}

Cond. X_{fer} = Com Total $I/p - IT$



$$a_{(auto)} = \frac{2250}{250} \\ = 9$$

$$S_{auto} = 2250 \times 10 \text{ (OR)} 250 \times 90 \\ = 22.5 \text{ kVA}$$

Conductive = -2.5 kVA

(worst option)

$$a(\text{auto}) \approx 1 \rightarrow S(\text{auto}) = [a(2\text{wdg}) \pm 1] \times S(2\text{wdg})$$

$$a(\text{auto}) > 1 \rightarrow S(\text{auto}) = \left[1 \pm \frac{1}{a(\text{auto})} \right] \times S(2\text{wdg})$$

- ⊗ An auto TF is a TF in which a part of wdg is common to 1° & 2° ckt's both.
- ⊗ Unlike a 2-wdg TF where there is only inductive X_{fer} , an auto TF provide inductive X_{fer} as well as conductive X_{fer} .
- ⊗ As compare to a 2-wdg TF For the same duty an auto TF uses less cu & less iron.
Its exciting current is lower & η is higher.
- ⊗ Its pu impedance & therefore its V_R is lower as compare to the 2wdg. X_{mer} .
However a lower pu impedance results into higher SC currents.
- ⊗ The cu saving is directly related to the ratio of conductive X_{fer} to total X_{fer} & in fact equals this ratio.
Hence auto TF are usually preferred when the vol. Ratio as auto TF is close to 1.
- ⊗ Practically therefore auto TFs are used where Voltage Ratio is limited to 2.
- ⊗ In exceptional circumstances Vol. Ratio of 3:1 may be used.
- ⊗ If a 2wdg TF has to be reconnected as auto TF then the insulation of its LV wdg should be strengthen to with stand the HV expected during operation as auto TF.
- ⊗ Auto TF has the disad. that it is not provide electrical isolation between the 2 ckt's.
And therefore the disturbance on one side may quickly spread to other side.

Also if in the stepdown modes, the common wdg develops an oc then the load may be subjected to the full HV of the source limited only by the vol. drop across the series wdg.

Applications →

(1) To connect 2 power sys. at diff. vol. levels where the vol. ratio is close to 1 & usually limited to 2.

eg:- 765kV/400kV, 440/220kV & 220kV/132kV etc.

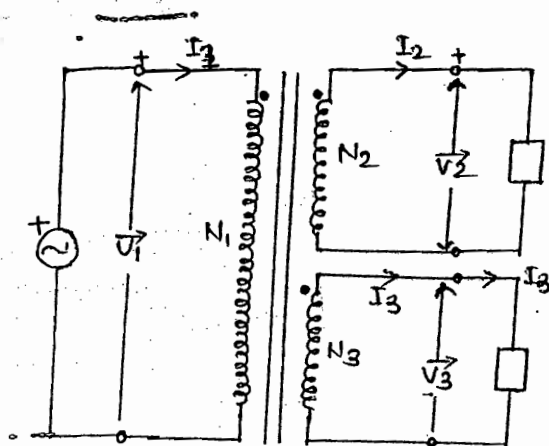
(2) As Booster for line drop compensation in electric traction supply systems.

(3) To start large 3- ϕ induction motors usually th of the squirrel cage type.

(4) In manual, automatic & servo-voltage stabilizers for domestic, commercial & industrial use.

AS continuously ^{variable} vol. TF (or) Lab. applications (variag)

primary winding →



$$\frac{V_1}{N_1} = \frac{V_2}{N_2} = \frac{V_3}{N_3}$$

mmf balance in load →

$$N_1 \vec{I}_1 - N_2 \vec{I}_2 - N_3 \vec{I}_3 = N_1 \vec{I}_0$$

$$\vec{I}_1 = \frac{N_2}{N_1} \times \vec{I}_2 + \frac{N_3}{N_1} \times \vec{I}_3 + \vec{I}_0$$

Taking conjugate

$$\vec{I}_1^* = \frac{N_2}{N_1} \times \vec{I}_2^* + \frac{N_3}{N_1} \times \vec{I}_3^* + \vec{I}_0^*$$

multiplying by $\vec{V}_1$

$$\vec{V}_1 \vec{I}_1^* = \left(\vec{V}_1 \times \frac{N_2}{N_1} \right) \times \vec{I}_2^* + \left(\vec{V}_1 \times \frac{N_3}{N_1} \right) \times \vec{I}_3^* + \vec{V}_1 \vec{I}_0^*$$

$$\vec{V}_1 \vec{I}_1^* = \vec{V}_2 \vec{I}_2^* + \vec{V}_3 \vec{I}_3^* + \vec{V}_1 \vec{I}_0^*$$

$$\vec{S}_1 = \vec{S}_2 + \vec{S}_3 + \vec{S}_0$$

Que. → The ratio of the no. of turns per phase in 1°, 2° & 3° wdg. of TF is 10:2:1 with lagging currents of 45A with PF 0.8 in the 2° & 50A at PF 0.71 in the 3° wdg, find the 1° current & PF. (Neglect losses & exciting currents)

soln →

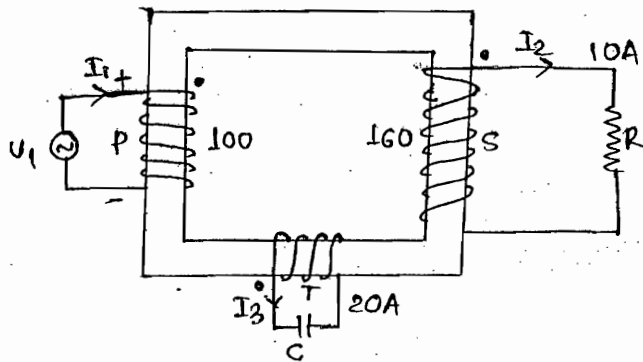
$$\vec{I}_1 = \frac{N_2}{N_1} \vec{I}_2 + \frac{N_3}{N_1} \times \vec{I}_3$$

$$= \frac{2}{10} \times 45 \angle \cos^{-1} 0.8 + \frac{1}{10} \times 50 \angle \cos^{-1} (0.71)$$

$$= 13.97 \angle -39.69^\circ \text{ A}$$

$$1^\circ \text{ PF} = \cos(39.69) = 0.7695 \text{ (lagging)}$$

Que →



An ideal TF has 3 wdg :-

- 100 turns on 1^o wdg P, 160 turn on 2^o wdg S & 60 turns on 3^o T.
- Wdg S feeds 10A to a resistive load whereas a pure capacitive load across wdg T takes 20A.

- Calculate the current in 1^o wdg & its PF in case TF magnetising current is neglected.
- With the polarity marking on P as shown mark the polarity on wdg S & T also.

Solⁿ → Taking vol. V_1 as ref vol.

$$I_1 = \frac{N_2}{N_1} I_2 + \frac{N_3}{-N_1} I_3$$

$$= \frac{160}{100} \times 10 + \frac{60}{100} \times 20 \angle 90^\circ$$

$$I_1 = 20 \angle 36.869^\circ \text{ A}$$

$$1^\circ \text{ PF} = \cos(36.869)$$

$$= 0.8 (\text{leading})$$

C → leading VARs
supply - leading

DATE: 02/11/19

Tertiary wdg (CONTD) →

- * It is the 3rd wdg in addition to usual 1^o & 2^o wdg.
- * It is provided to give a 3rd vol. level for the special requirements such as :-
 - (1) To interconnect 3 power sys. at different vol. levels.
 - (2) To provide 2 diff. auxiliary vol. levels in unit auxiliary TF of power substations.
 - (3) To connect reactive power compensating equipment in substations.
 - (4) To provide low vol. supply for water, lightning, air conditioning & heating, (or) operating personal in high vol. substations.
- * A tertiary delta wdg is provided in star by star TF to overcome prob. related to magnetising current phenomenon; unbalanced load & unbalanced faults.
- Such a 3^oΔ wdg if unloaded is called stabilising wdg.